

Waves reflect when $k_r V_A \sim \omega$

CZ shuts off when $V_A \sim c_s \lambda$, $\lambda \equiv V_R - V_A$

$$\left(\frac{V_A^c}{V_A^w} \right)^2 = \left(\frac{c_s \lambda k_r}{\omega} \right)^2 = \left(\frac{c_s \lambda k_e N}{\omega^2} \right)^2 > \frac{c_s^2 \lambda^2 k_e^2 N^2}{\omega_m^4}$$

$$k_r = k_e \frac{N}{\omega}$$

$$L_d = \frac{2 \omega^4}{k_e^3 N N_T^2 \alpha} = r, N \approx N_T$$

$$\rightarrow \omega_{min}^2 = \sqrt{\frac{\alpha r N N_T^2 k_e^3}{2}}$$

$$\left(\frac{V_A^c}{V_A^w} \right)^2 > \frac{2 c_s^2 \lambda^2 k_e^2 N^2}{\alpha r N^3 k_e^3} = \frac{2 c_s^2 \lambda^2}{\alpha r N k_e} \sim \frac{2 c_s^2 \lambda^2}{\alpha N \ell}$$

$$= \frac{2C_s^2 \lambda^7}{N\ell} \left(\frac{16\sigma T^3}{3kg^2 C_p} \right)^{-1}$$

$$= \frac{3kg^2 C_p C_s^2 \lambda^7}{8\sigma T^3 N\ell} = \frac{g^2 g \sigma T^4 V_R C_p C_s^2 \lambda^7}{8 P F \sigma T^3 N\ell}$$

$$\begin{aligned} 3k &= \frac{4\pi\sigma g r^2 T^4}{PL} V_R \\ &= \frac{\sigma g T^4 V_R}{PF} \end{aligned} \quad \left| \begin{aligned} &= \frac{g^2 g C_p T C_s^2 \lambda^7 V_R}{8 P F N\ell} \\ &\approx \frac{P g \lambda^7 V_R}{8 F N\ell} \\ &= \frac{1}{8\ell} V_R \lambda^7 \left(\frac{gP}{FN} \right) \end{aligned} \right.$$

$$\approx \frac{1}{8\ell} V_R \left(\frac{N h P}{F} \right)$$

$$\lambda \approx \frac{N^2 h}{g}$$

$$\approx \frac{1}{8\ell} V_R \left(\frac{N h}{c} \right) \left(\frac{P}{P_{rad}} \right)$$

$$\approx \frac{1}{8\ell} V_R \sqrt{\lambda} \left(\frac{C_s}{c} \right) \left(\frac{P}{P_{rad}} \right)$$